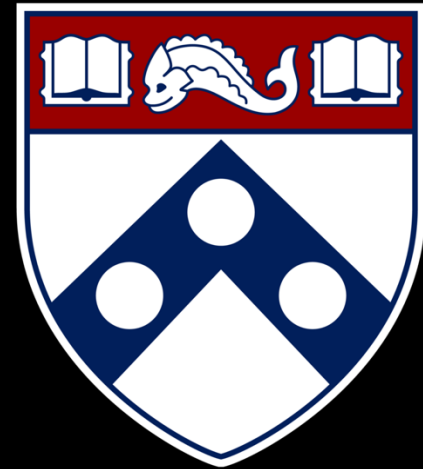


Empirical Security & Privacy, for Humans



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Privacy and Human Behavior in the Age of Information

Privacy and human behavior in the age of information

Alessandro Acquisti,^{1*} Laura Brandimarte,¹ George Loewenstein²

This Review summarizes and draws connections between diverse streams of empirical research on privacy behavior. We use three themes to connect insights from social and behavioral sciences: people's uncertainty about the consequences of privacy-related behaviors and their own preferences over those consequences; the context-dependence of people's concern, or lack thereof, about privacy; and the degree to which privacy concerns are malleable—manipulable by commercial and governmental interests. Organizing our discussion by these themes, we offer observations concerning the role of public policy in the protection of privacy in the information age.

If this is the age of information, then privacy is the issue of our times. Activities that were once private or shared with the few now leave trails of data that expose our interests, traits, beliefs, and intentions. We communicate using e-mails, texts, and social media; find partners on dating sites; learn via online courses; seek responses to mundane and sensitive questions using search engines; read news and books in the cloud; navigate streets with geotracking systems; and celebrate our newborns, and mourn our dead, on social media profiles. Through these and other activities, we reveal information—both knowingly and unwittingly—to one another, to commercial entities, and to our governments. The monitoring of personal information is ubiquitous; its storage is so durable as to render one's past undeletable (1)—a modern digital skeleton in the closet. Accompanying the acceleration in data collection are steady advancements in the ability to aggregate, analyze, and draw sensitive inferences from individuals' data (2).

Both firms and individuals can benefit from the sharing of once hidden data and from the application of increasingly sophisticated analytics to larger and more interconnected databases (3). So too can society as a whole—for instance, when electronic medical records are combined to observe novel drug interactions (4). On the other hand, the potential for personal data to be abused—for economic and social discrimination, hidden influence and manipulation, coercion, or censorship—is alarming. The erosion of privacy can threaten our autonomy, not merely as consumers but as citizens (5). Sharing more personal data does not necessarily always translate into more progress, efficiency, or equality (6).

Because of the seismic nature of these developments, there has been considerable debate about individuals' ability to navigate a rapidly evolving privacy landscape, and about what, if anything, should be done about privacy at a policy level. Some trust people's ability to make self-interested

decisions about information disclosing and withholding. Those holding this view tend to see regulatory protection of privacy as interfering with the fundamentally benign trajectory of information technologies and the benefits such technologies may unlock (7). Others are concerned about the ability of individuals to manage privacy amid increasingly complex trade-offs. Traditional tools for privacy decision-making such as choice and consent, according to this perspective, no longer provide adequate protection (8). Instead of individual responsibility, regulatory intervention may be needed to balance the interests of the subjects of data against the power of commercial entities and governments holding that data.

Are individuals up to the challenge of navigating privacy in the information age? To address this question, we review diverse streams of empirical privacy research from the social and behavioral sciences. We highlight factors that influence decisions to protect or surrender privacy and how, in turn, privacy protections or violations affect people's behavior. Information technologies have progressively encroached on every aspect of our personal and professional lives. Thus, the problem of control over personal data has become inextricably linked to problems of personal choice, autonomy, and socioeconomic power. Accordingly, this Review focuses on the concept of, and literature around, informational privacy (that is, privacy of personal data) but also touches on other conceptions of privacy, such as anonymity or seclusion. Such notions all ultimately relate to the permeable yet pivotal boundaries between public and private (9).

We use three themes to organize and draw connections between streams of privacy research that, in many cases, have unfolded independently. The first theme is people's uncertainty about the nature of privacy trade-offs, and their own preferences over them. The second is the powerful context-dependence of privacy preferences: The same person can in some situations be oblivious to, but in other situations be acutely concerned about, issues of privacy. The third theme is the malleability of privacy preferences, by which we mean that privacy preferences are subject to

influence by those possessing greater insight into their determinants. Although most individuals are probably unaware of the diverse influences on their concern about privacy, entities whose interests depend on information revelation by others are not. The manipulation of subtle factors that activate or suppress privacy concern can be seen in myriad realms—such as the choice of sharing defaults on social networks, or the provision of greater control on social media—which creates an illusion of safety and encourages greater sharing.

Uncertainty, context-dependence, and malleability are closely connected. Context-dependence is amplified by uncertainty. Because people are often “at sea” when it comes to the consequences of, and their feelings about, privacy, they cast around for cues to guide their behavior. Privacy preferences and behaviors are, in turn, malleable and subject to influence in large part because they are context-dependent and because those with an interest in information divulgence are able to manipulate context to their advantage.

Uncertainty

Individuals manage the boundaries between their private and public spheres in numerous ways: via separateness, reserve, or anonymity (10); by protecting personal information; but also through deception and dissimulation (11). People establish such boundaries for many reasons, including the need for intimacy and psychological respite and the desire for protection from social influence and control (12). Sometimes, these motivations are so visceral and primal that privacy-seeking behavior emerges swiftly and naturally. This is often the case when physical privacy is intruded—such as when a stranger encroaches in one's personal space (13–15) or demonstratively eavesdrops on a conversation. However, at other times (often including when informational privacy is at stake) people experience considerable uncertainty about whether, and to what degree, they should be concerned about privacy.

A first and most obvious source of privacy uncertainty arises from incomplete and asymmetric information. Advancements in information technology have made the collection and usage of personal data often invisible. As a result, individuals rarely have clear knowledge of what information other people, firms, and governments have about them or how that information is used and with what consequences. To the extent that people lack such information, or are aware of their ignorance, they are likely to be uncertain about how much information to share.

Two factors exacerbate the difficulty of ascertaining the potential consequences of privacy behavior. First, whereas some privacy harms are tangible, such as the financial costs associated with identity theft, many others, such as having strangers become aware of one's life history, are intangible. Second, privacy is rarely an unalloyed good; it typically involves trade-offs (16). For example, ensuring the privacy of a consumer's

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Alan Westin Privacy Segmentation Index (1995)

Questions:

How strongly do you agree or disagree?

- A. Consumers have lost all control over how personal information is collected and used by companies.
- B. Most businesses handle the personal information they collect about consumers in a proper and confidential way.
- C. Existing laws and organizational practices provide a reasonable level of protection for consumer privacy today.

PC?

3,4

1,2

1,2

1 = Strongly Disagree,

2 = Somewhat

Disagree,

3 = Somewhat Agree,

4 = Strongly Agree

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Segment

I. Privacy Fundamentalists:

Participants who give privacy concerned responses to **all questions**;

II. Privacy Unconcerned:

Participants who give privacy concerned responses to **no questions**;

III. Privacy Pragmatists: All other participants

1996

Res

25%

16%

59%

1 = Strongly Disagree, 2 = Somewhat Disagree,
3 = Somewhat Agree, 4 = Strongly Agree

The Privacy Paradox (2007)

People say they care about privacy. So why do they share so much?



Those who bucket as Westin's fundamentalists regularly share sensitive data for small rewards

100

THE JOURNAL OF CONSUMER AFFAIRS

PATRICIA A. NORBERG, DANIEL R. HORNE,
AND DAVID A. HORNE

The Privacy Paradox: Personal Information Disclosure Intentions versus Behaviors

Impelled by the development of technologies that facilitate collection, distribution, storage, and manipulation of personal consumer information, privacy has become a "hot" topic for policy makers. Commercial interests seek to maximize and then leverage the value of consumer information, while, at the same time, consumers voice concerns that their rights and ability to control their personal information in the marketplace are being violated. However, despite the complaints, it appears that consumers freely provide personal data. This research explores

The Privacy Paradox

People say they care about privacy. So why do they share so much?



Benefits

- Personalization
- Convenience
- Social connection

But also risks

- Discrimination
- Manipulation
- Loss of autonomy

Why? Answer in three themes

Uncertainty

- Don't know consequences of sharing
- Don't know own preferences

Context-dependence

- Can be privacy-obsessed or careless; depends on situation
- Environmental cues a strong influence on behavior

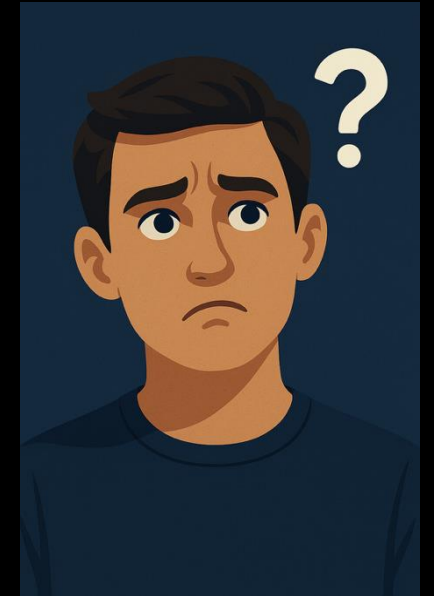
Malleability

- Privacy preferences can be manipulated
- Those who profit from data sharing can exploit psychological processes

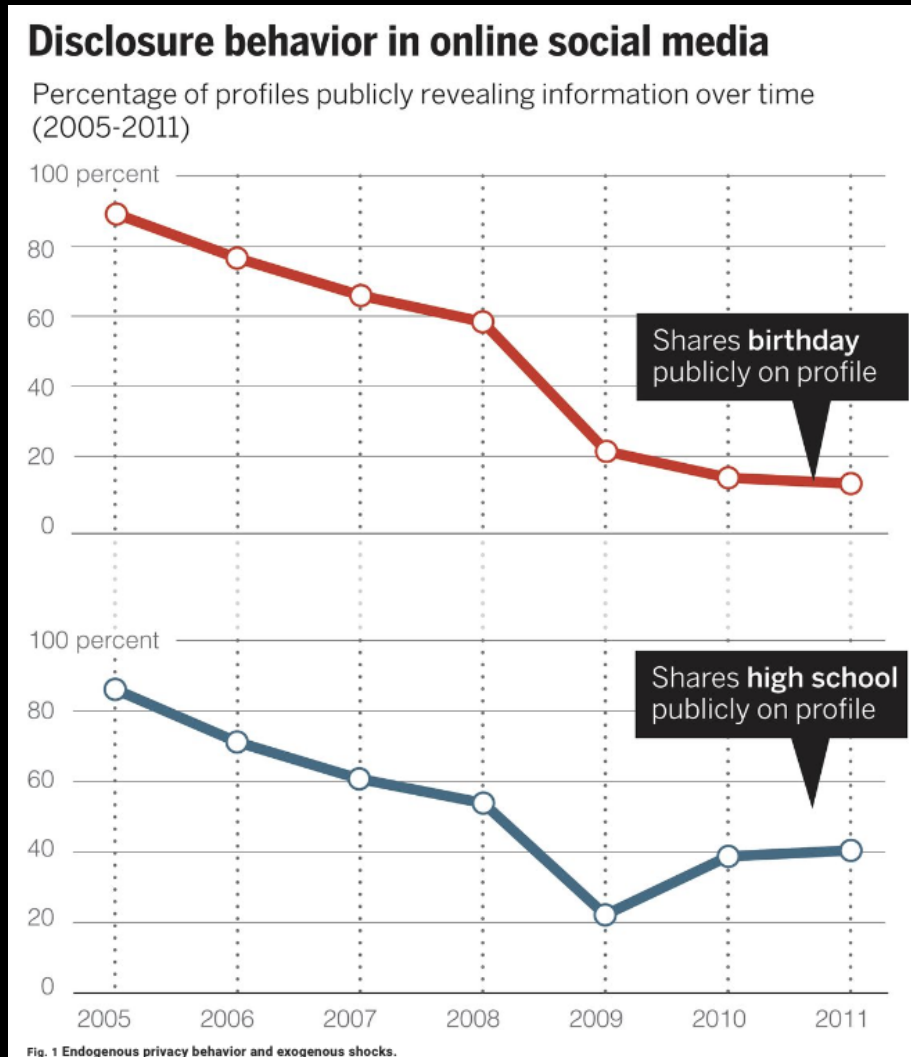


Uncertainty

- Don't know consequences of sharing
 - Don't know own preferences
-
- When consequences are tangible and clear, people act on privacy concerns
 - But:
 - Immediate gratification > delayed consequences
 - Misperceptions of actual costs & benefits
 - Mental shortcuts, emotional factors
 - Fundamental tension: Need for privacy vs. desire to share and connect



Context-dependence



- Can be privacy-obsessed or careless; depends on situation

Context-dependence

- Can be privacy-obsessed or careless; depends on situation
- Environmental cues are a strong influence

Disclosure behavior in online social media

Percentage of profiles publicly revealing information over time (2005-2011)

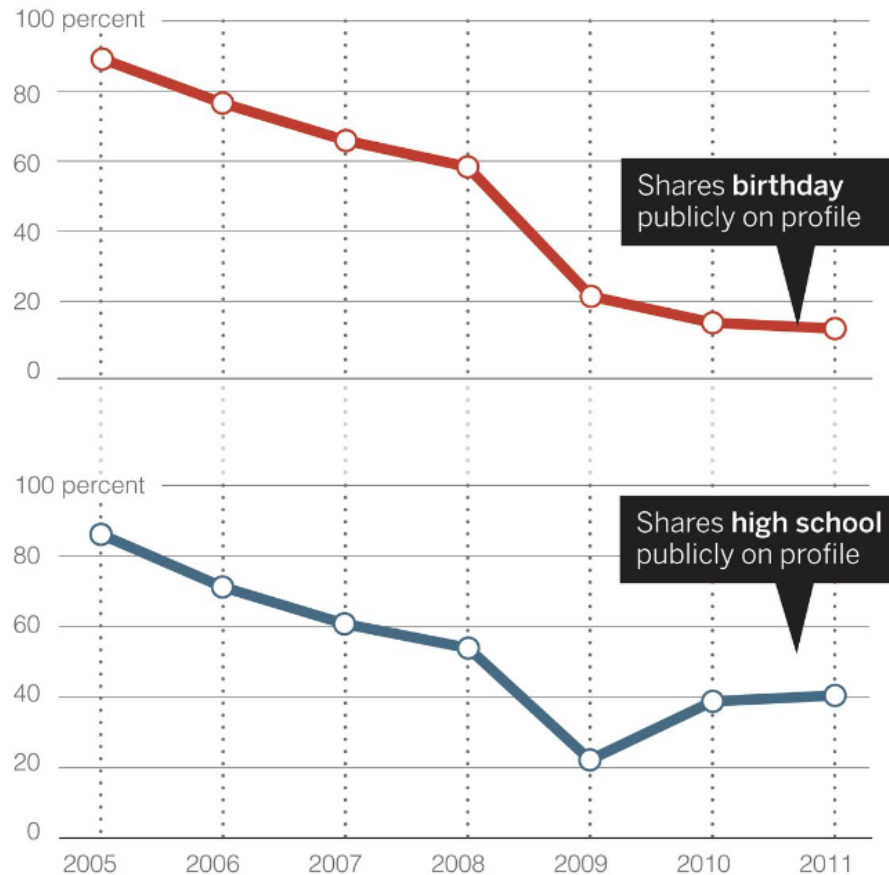
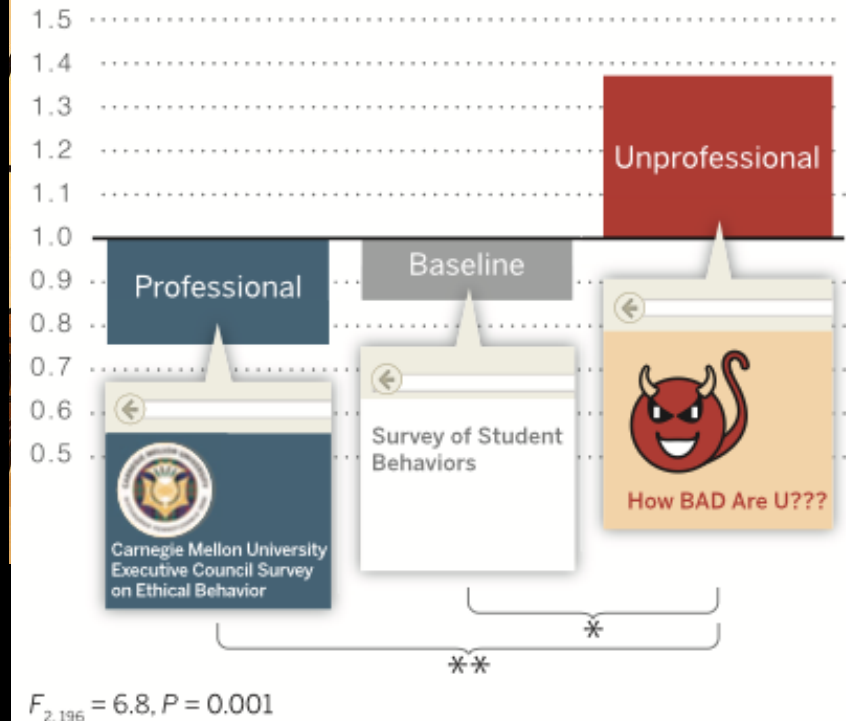


Fig. 1 Endogenous privacy behavior and exogenous shocks.



A measure of privacy behavior

Relative admission rates in an experiment testing the impact of different survey interfaces on willingness to answer questions about various sensitive behaviors

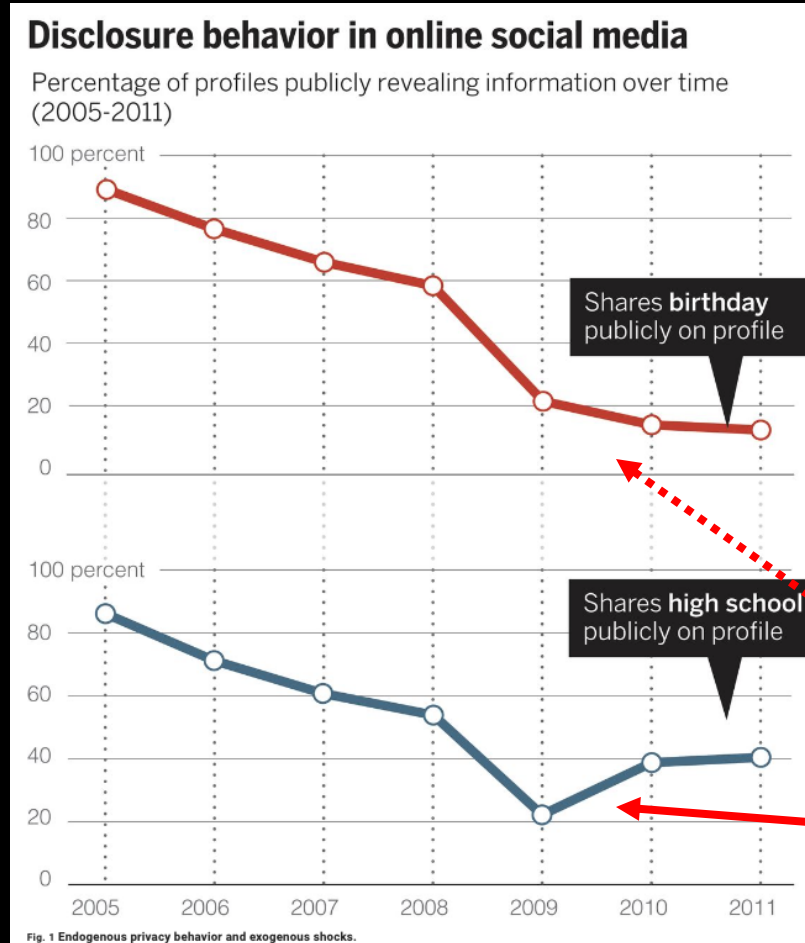


Context-dependence: Other factors

- Social cues
 - Imitation (“majority admitted this behavior”)
 - Reciprocity
- Adaptation over time
 - Surveillance (“breaking ice” effect)
 - Change activates concern; stability dulls it
- On-line setting: Context confusion
 - Boundaries are blurry
 - Privacy management both more complex and more consequential

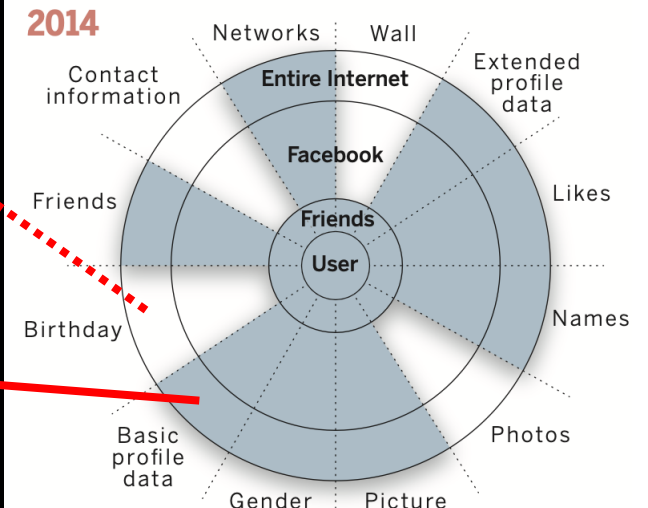
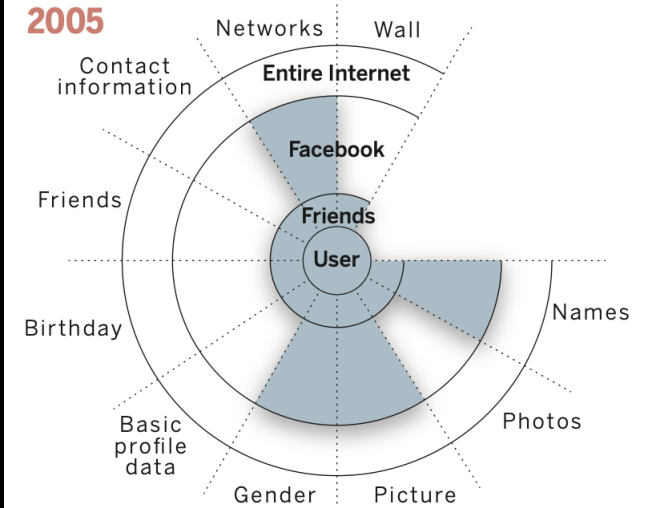
Malleability

- Privacy preferences can be manipulated
- Those who profit from data sharing can exploit psychological processes



Default visibility settings in social media over time

Visible (default setting) Not visible



Malleability: Other factors

- The paradox of trust and control
 - Simply posting unread privacy policy → misplaced feelings of protection
 - Greater control over disclosure → reduced concern → more sharing
- Privacy policies
 - Most people don't read them – if they did, est. \$781B prod. cost/year !
 - Transparency requires comprehensibility, accessibility, actionability
- Not all malleability is sinister
 - Monitoring, good defaults, simple cues can increase prosocial behavior
 - Anonymity can increase truthful responses but also enable cheating and dishonesty

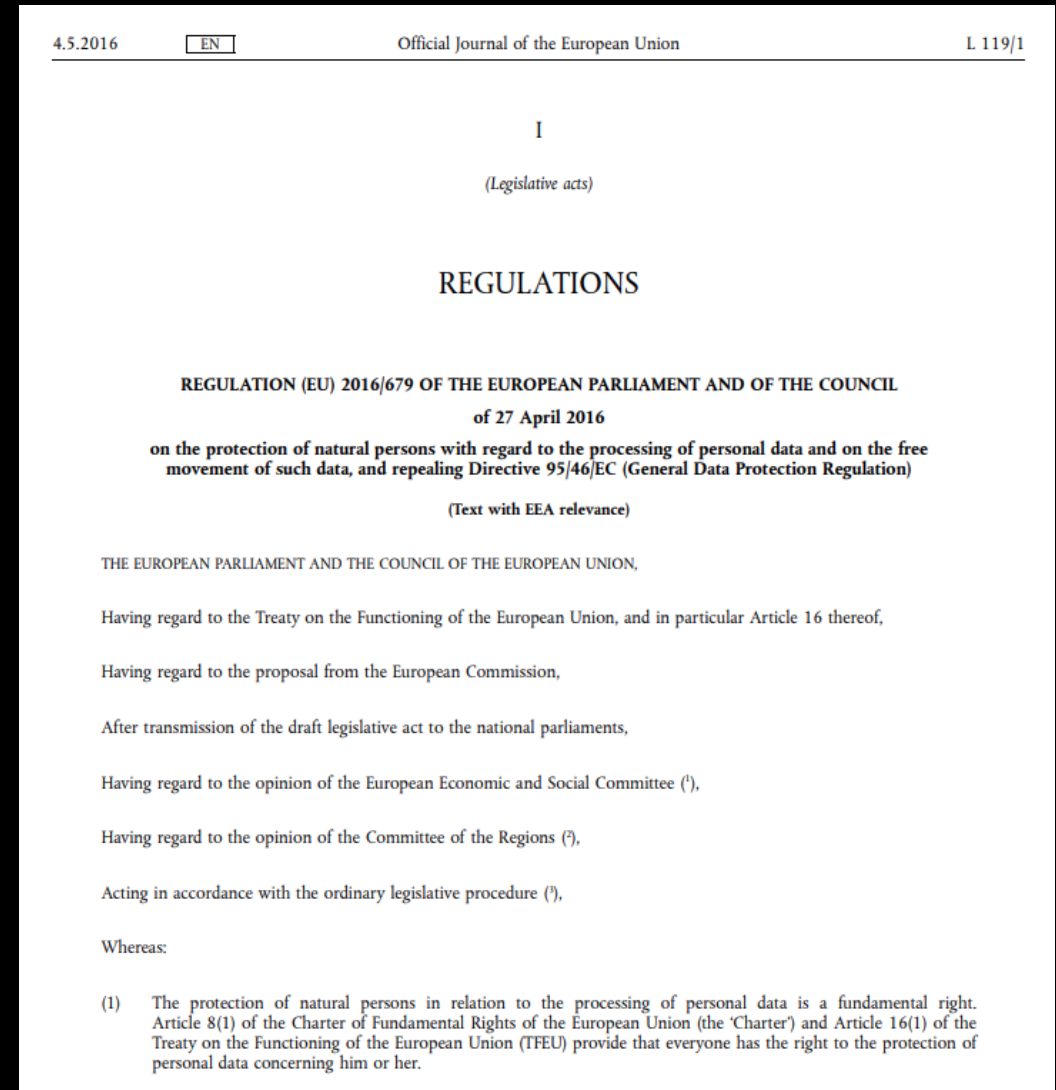
Policy implications

- People cannot adequately self-protect
 - So individual empowerment is insufficient
- Need baseline protection framework
 - Fair information principles and practices
 - Protection that doesn't require constant informed decision-making
 - Accommodate real humans: "naïve, uncertain, and vulnerable"
 - **Goal: Equity of power**



General Data Protection Regulation (GDPR)

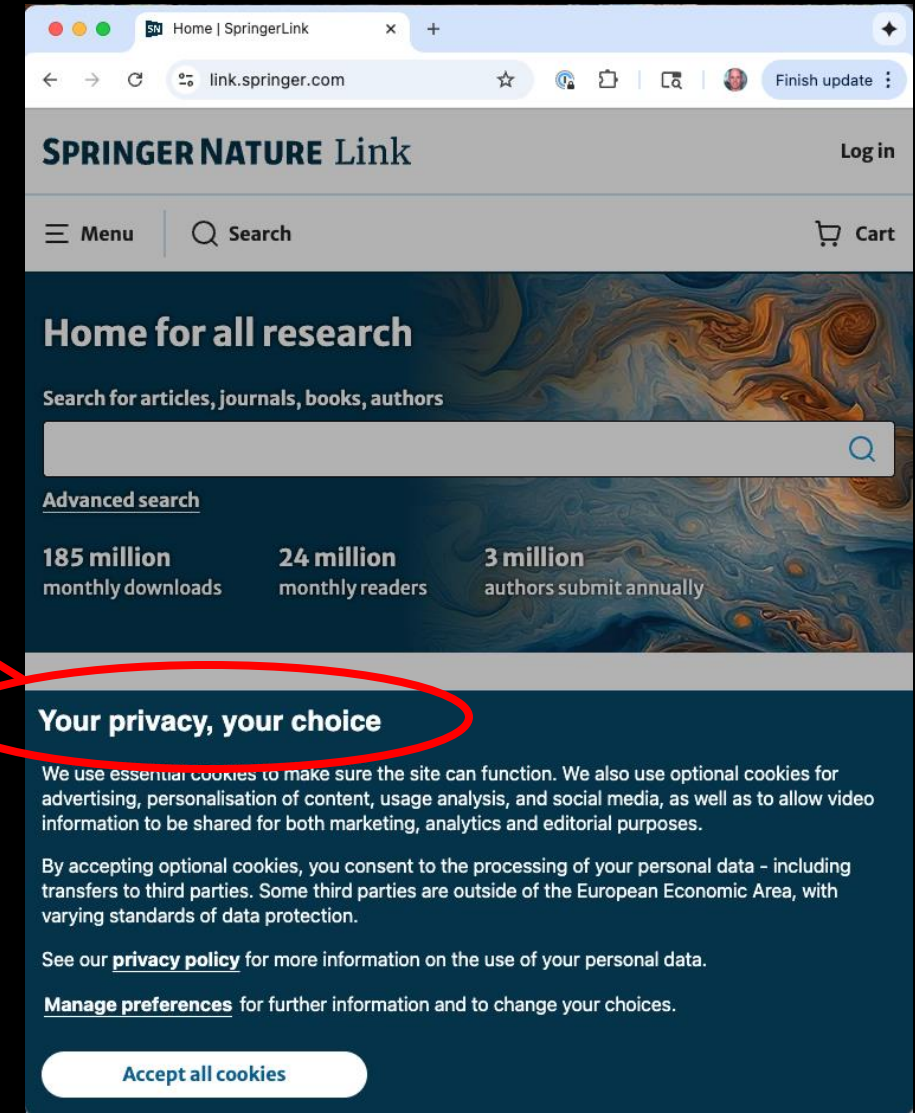
- Regulates and protects the processing of personal information (of EU citizens)
- Companies must be transparent about the personal data they handle and have a legitimate purpose for using it



GDPR: Personal rights

Data subjects have a Right

- to be Informed
- of Access
- to Rectification
- to Erasure
- to Restrict Processing
- to Data Portability
- to Object
- to not to be subject to a decision based solely on automated processing (with exceptions)



GDPR and context-dependence, malleability

Context dependence

- **Habituation:** Repeated consent warnings cause fatigue, leading to surveillance acquiescence
- **Control:** Fine-grained data controls encouraged by the law, but paradoxically discourage privacy

Malleability

- **Consent:** Using "pre-ticked boxes" or convoluted consent forms that aren't freely given
- **Transparency:** Making privacy policies overly complex or hard to find, despite GDPR requiring transparency

Law does not force companies to reveal **uncertain** benefits and consequences, nor require them to **nudge** people toward the former

Key takeaways

- **The privacy paradox is real** → people say one thing, do another
- **Three interconnected themes** explain the paradox:
 - Uncertainty amplifies context-dependence
 - Context-dependence enables malleability
 - Malleability creates power imbalances
- **Our instincts are inadequate** for navigating information age privacy
- **Policy must protect vulnerable humans**, not assume rational actors
- **Baseline protections are necessary**, not just individual choice

Discussion questions

- Do you recognize these patterns in your own privacy behavior?
- What examples of "malicious design" have you encountered?
- Should there be stronger regulations on default settings?
- How do we balance innovation benefits with privacy protection?
- Is the "notice and consent" model salvageable, or do we need a completely different approach?